MIRAMAR FORKLIFT

OSHA-COMPLIANT CERTIFICATION

Miramar Forklift Training

Professional Powered Industrial Truck (PIT) Operator Certification — meeting all applicable OSHA standards. This training is designed to equip operators with the knowledge, skills, and judgment needed to work safely and effectively in any warehouse or industrial environment.





Why This Training Matters

85–100

Annual Fatalities

Workers killed each year in U.S. manufacturing due to Powered Industrial Truck accidents.

95K

Annual Injuries

Forklift-related incidents result in lost-time accidents across all industries every year.

~70%

Preventable

OSHA estimates the vast majority of forklift accidents are directly linked to improper or inadequate operator training.

OSHA Finding: The majority of forklift incidents — injury, property damage, or death — are directly linked to **inadequate or improper training**. Proper certification is not optional; it is the law.



REGULATORY FRAMEWORK

OSHA Guidelines That Apply

Forklift operator training is governed by multiple OSHA standards depending on the industry and worksite. The following regulations establish the legal requirements for PIT operator certification:

General Industry

29 CFR **1910.178(l)** — Powered Industrial Trucks

Shipyard Employment

29 CFR **1915.120(a)**

Marine Terminals

29 CFR **1917.1(a)(2)(xiv)**

Longshoring

29 CFR **1918.1(b)(10)**

Construction

29 CFR **1926.602(d)**

CERTIFICATION PROCESS

OSHA-Compliant Certification Steps

Review Rules
Read OSHA safe operating rules

Watch Video
View PIT operation demonstration

Written Exam
Complete knowledge assessment

Hands-On
Class IV forklift equipment familiarization

Completion of all four steps is required to receive your Miramar Forklift Training Certificate of Achievement and your Certified Operator Card. Your certification remains valid for **three years**.



It's the Law

To operate a Powered Industrial Truck (PIT) or forklift legally in the United States, you must meet the following requirements set by federal law.

Minimum Age

You must be at least **18 years of age** to be certified as a PIT operator.

Authorized Personnel Only

Only **trained and authorized personnel** are permitted to operate Powered Industrial Trucks.

Recertification Required

Certification must be renewed every **three years** or following any observed unsafe operation.



Upon completing all certification requirements, operators receive a Miramar Forklift Training Certificate of Achievement and a Certified Operator Card.



CHAPTER 1

Equipment Types & OSHA Classifications

Understanding the different categories of Powered Industrial Trucks is foundational to safe operation. OSHA recognizes **7 classifications** of PITs, each with distinct design characteristics, fuel types, and recommended operating environments.

Upon certification through Miramar Forklift Training, you will be licensed to operate all seven PIT classifications listed in this section.

What Is a Powered Industrial Truck?

A **Powered Industrial Truck (PIT)** is any mobile, self-propelled vehicle used to carry, push, pull, lift, stack, or tier materials. These vehicles are commonly known as:

- Forklifts & Fork Trucks
- Pallet Jacks & Rider Trucks
- Lift Trucks

PITs may be powered by **electric motors** or **internal combustion engines**. This definition excludes earth-moving equipment and over-the-road hauling vehicles.





CLASS I

Electric Motor Rider Trucks

Class I trucks are battery-powered counterbalanced forklifts designed for both indoor and outdoor use. They produce zero direct emissions, making them ideal for enclosed environments with limited ventilation.

Solid / Cushion Tires

Best suited for smooth, hard surfaces such as indoor warehouse floors.

Pneumatic Tires

Designed for slightly rougher surfaces while still maintaining a clean power source.

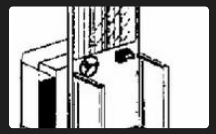
CLASS II

Electric Motor Narrow Aisle Trucks



Reach Truck

Extendable mast reaches deep into racking systems for high-density storage retrieval.



Order Picker

Elevates the operator to pick items directly from high shelving. Operator travels with the platform.



Turret Truck

Features a 90-degree rotating turret, allowing forks to pick from either side of a very narrow aisle.

All Class II trucks use **solid/cushion tires** and are optimized for high-density indoor warehouse environments where aisle space is at a premium.

CLASS III

Walkie & Rider Pallet Jacks & Stackers



Class III Key Features

- Operator walks behind or rides on the vehicle
- Includes walkie pallet jacks, walkie stackers, and rider reach trucks
- Solid/cushion tires only — designed for hard, even surfaces
- Electric-powered — zero direct emissions
- Ideal for dock-to-floor and low-height stacking operations



CLASS IV

IC Rider Trucks — Cushion Tire

Class IV trucks are internal combustion (IC) engine forklifts equipped with **solid or cushion tires**, designed primarily for smooth indoor surfaces such as concrete warehouse floors. This is the class covered in depth during Miramar's hands-on certification training.

Liquid Propane (LPG)

Most common fuel for indoor IC forklifts. Cleaner burning than diesel or gasoline.

Gasoline

Higher emissions; less common indoors. Used in applications with adequate ventilation.

Diesel

High power output; typically reserved for heavier loads and outdoor-adjacent environments.

CLASS V

IC Engine Trucks — Pneumatic Tire

Class V Overview

Class V trucks use internal combustion engines paired with **pneumatic tires**, making them well-suited for outdoor use and uneven terrain. Common fuel types include LPG, gasoline, and diesel.

- Pneumatic tires for outdoor and semi-rough surfaces
- Higher load capacities than Class IV models
- Commonly used in lumber yards, construction sites, and outdoor storage
- Ventilation requirements apply for indoor use due to IC engine emissions



CLASS VI

IC Engine Tractors



Class VI — Towing Vehicles

Class VI trucks are designed for **towing or pushing (not lifting)** heavy loads. They are the primary PIT used at airports, large distribution centers, and manufacturing facilities where loads travel on wheeled carts.

- Air pneumatic tires for outdoor paved surfaces
- IC engine power — gasoline, diesel, or LPG
- Used extensively in airport ground operations and large campus logistics
- Load capacity measured in pull force, not lift weight

CLASS VII

Rough Terrain Forklift Trucks



Telescopic Boom

Extendable boom arm for reaching across or over obstacles. Common on construction sites.



Vertical Mast

Traditional mast design on large, knobby tires. Used for outdoor yards, lumber, and agriculture.

Class VII trucks are built for unimproved surfaces — packed gravel, dirt, mud, and grass. They feature large, knobby pneumatic tires and higher ground clearance than other PIT classes.

CHAPTER 2

Appendix A: Forklift Tire Types

Selecting the right tire type is critical for safe and efficient forklift operation. Using the wrong tire for the surface can lead to instability, reduced control, and premature wear. There are three main forklift tire types — each suited to specific operating environments.



Cushion Tires

Solid rubber pressed onto a steel band. Best for smooth, hard indoor surfaces like concrete and asphalt. Not suitable for outdoor or uneven terrain.



Air Pneumatic Tires

Air-filled tires with aggressive tread for traction on loose, packed gravel, or packed dirt. Susceptible to puncture; require regular pressure checks.



Solid Pneumatic Tires

Same tread and footprint as air pneumatic, but made of solid rubber. Puncture-proof and longer lasting — but at a higher upfront cost.

CHAPTER 3

Appendix B: Forklift Fuel Types



Electric Battery

Zero direct emissions during operation. Ideal for enclosed, poorly ventilated spaces. Produces minimal hydrogen during recharging — ensure adequate airflow at charging stations.



Liquid Propane (LPG)

Suitable for indoor/outdoor use with adequate airflow. Produces some gas fume emissions during operation. Most common IC fuel choice for warehouse environments.



Liquid Fuel (Gas / Diesel)

Best for high-capacity forklifts operating primarily outdoors. Highest emission levels — not suitable for enclosed spaces without robust ventilation.

CHAPTER 4

Appendix C: Forklift Attachment Types

Specialized attachments extend the capabilities of a standard forklift for unique material handling needs. Each attachment changes the load capacity and center of gravity — always verify the modified data plate before lifting.

Fork Positioner

Hydraulically adjusts fork spacing without leaving the cab. Improves efficiency and reduces operator strain.

Rotating Carriage

Rotates the entire load forward for dumping or pouring from heavy containers.

Tire Handler

Designed to lift, rotate, and transport large tires for heavy equipment maintenance.

Barrel Attachment

Clamps and lifts barrels or drums for controlled pouring and precise placement.

Hook Attachment

Enables lifting and transporting of suspended loads via a rated hook and chain.

Roll Clamp

Squeezes and lifts paper rolls, large barrels, and cylindrical loads without pallets.



Fork Positioner

Reposition forks hydraulically from the operator's seat to match various pallet sizes — no manual adjustment needed.



Rotating Carriage

Dumps or pours contents from large industrial containers. Requires extra care — load center shifts dramatically during rotation.



Roll Clamp

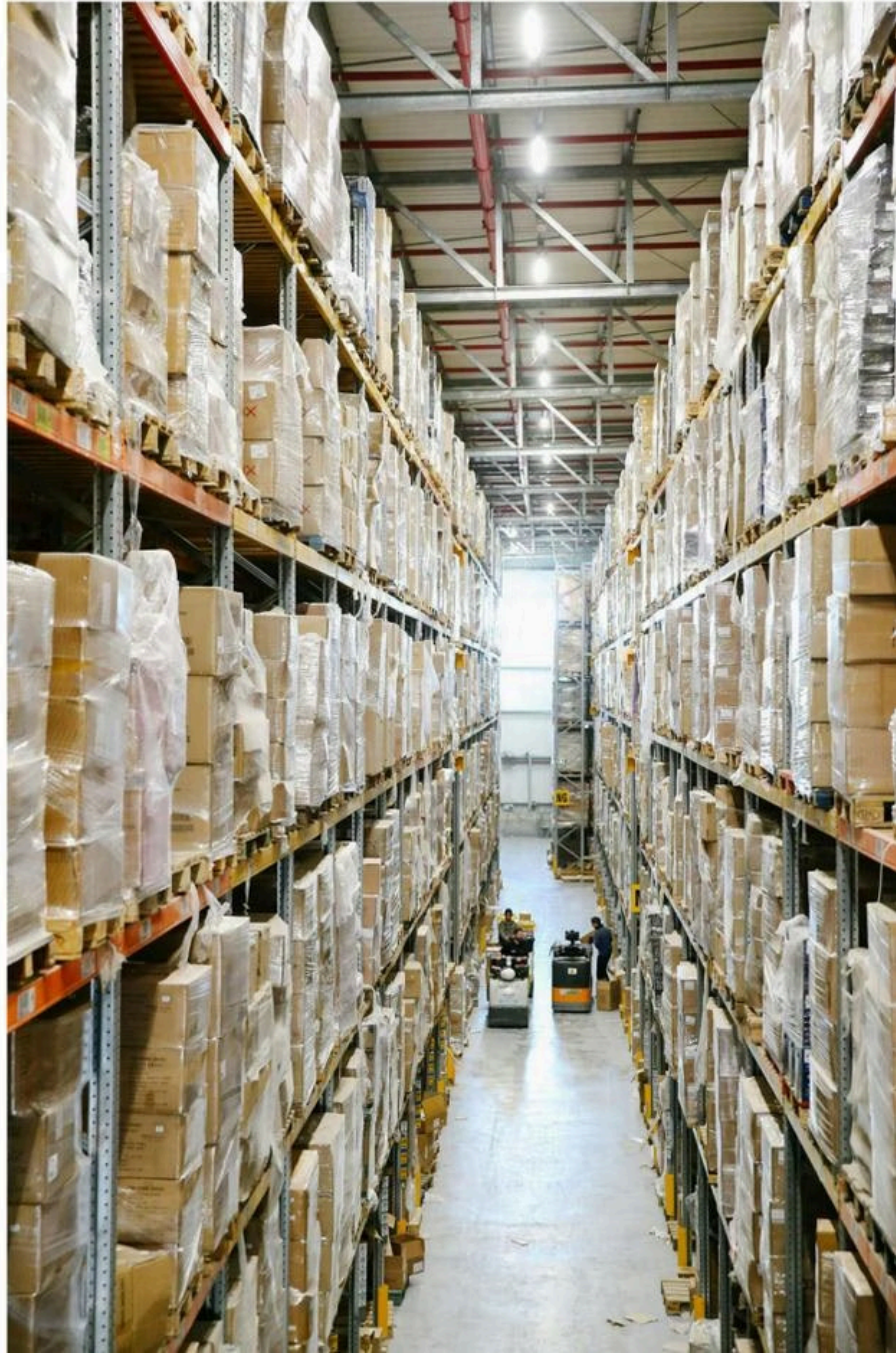
Grips large cylindrical loads such as paper rolls without a pallet. Rated for specific clamp force — never exceed the attachment's capacity.

Important: Any attachment changes the forklift's rated lifting capacity. Always check the **modified data plate** — never rely on the standard capacity when using an attachment.

CHAPTER 5

Occupational Safety & Operator Responsibility

OSHA — the Occupational Safety and Health Administration — is a federal agency established by law in 1970 and updated in 1999. Its mission: ensure safe and healthful working conditions for all U.S. workers through standards enforcement, training, outreach, and education. Every operator has a personal responsibility to follow OSHA standards on the job.



Who Is OSHA?

The **Occupational Safety and Health Administration** sets and enforces workplace safety standards across all U.S. industries.

OSHA's core mandate:

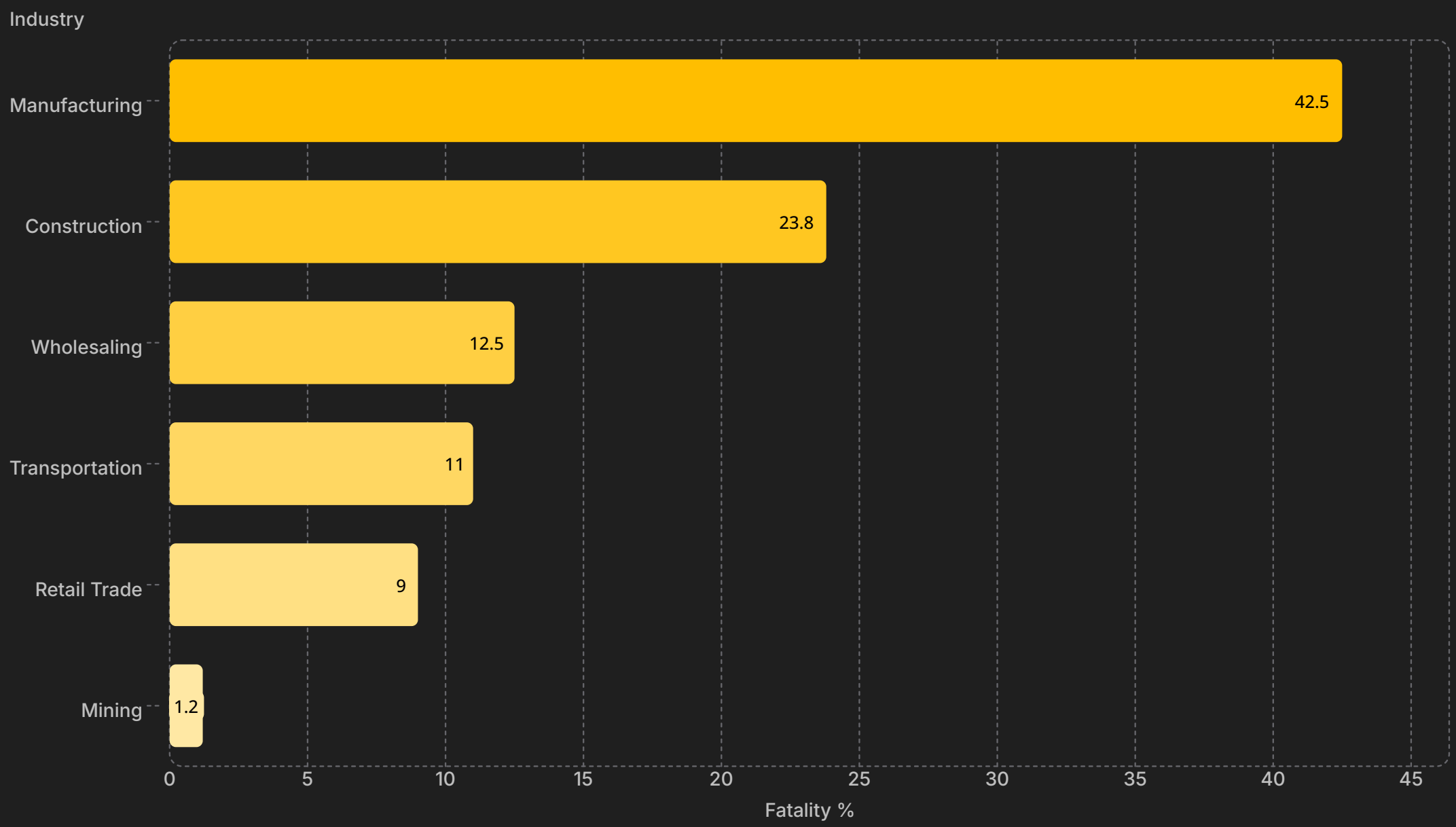
- Set enforceable safety standards
- Conduct unannounced site inspections
- Investigate workplace fatalities and injuries
- Provide training, outreach, and assistance



OSHA Requirement: Only **trained and authorized personnel** are permitted to operate Powered Industrial Trucks. This is non-negotiable.

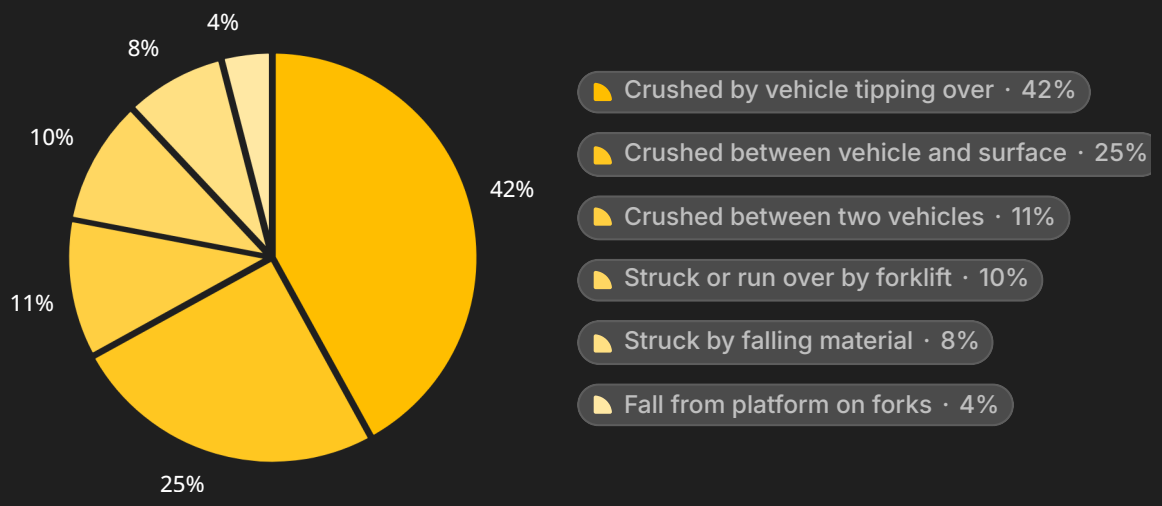
Forklift Fatalities By Industry

Forklift accidents can occur in any industry that uses powered industrial trucks. Manufacturing accounts for the largest share of fatalities, but no sector is immune. Understanding where accidents happen most frequently reinforces why proper training is a universal requirement.



Manufacturing leads at 42.5% of all forklift fatalities, followed by Construction at 23.8%. Together, these two industries account for nearly two-thirds of all PIT-related deaths annually.

How Forklift Fatal Accidents Happen



Key Insight

67% of all fatal forklift accidents involve the vehicle tipping over or the operator being crushed between the forklift and a fixed surface.

Both of these accident types are **directly preventable** through proper training, safe operating procedures, and maintaining the forklift's stability triangle.

The primary goal of forklift safety training is to **reduce accidents, injuries, and fatalities** in the workplace.

OSHA Site Inspections

Unannounced Inspections

As a federal and state agency, OSHA is authorized to conduct **unannounced inspections** at any time to verify compliance with workplace safety regulations.

Incident Investigations

OSHA investigates all incidents resulting in **serious injury or fatality** caused by workplace accidents, including forklift incidents.

Employee Responsibility

As an employee, you are **personally responsible** for strictly following the safety practices established by your employer and required by OSHA standards.



⚠ FINANCIAL RISK

The Cost of Non-Compliance

If OSHA determines that forklift operators are working without proper certification, employers face serious financial penalties.

$7K

Per Day, Per Employee

OSHA fines for each uncertified forklift operator — retroactive to the employee's hire date.

~$2M

Case Study Fine

If an operator has worked uncertified for one full year, the fine per employee could reach nearly **$2 million**.

Bottom Line: Certification is not just a safety best practice — it is a legal obligation with severe financial consequences for non-compliance.





Accidents Are Costly — In Every Way

Forklift accidents carry consequences far beyond the immediate physical harm. The human cost — injury, disability, or death — is irreversible. But the financial and reputational damage to a company can be equally devastating.

Human Cost

Injuries and fatalities affect operators, coworkers, and their families permanently.

Financial Cost

OSHA fines, legal liability, workers' compensation claims, and lost productivity add up quickly.

Reputational Cost

Workplace accidents attract media coverage and damage a company's standing with clients, partners, and future employees.



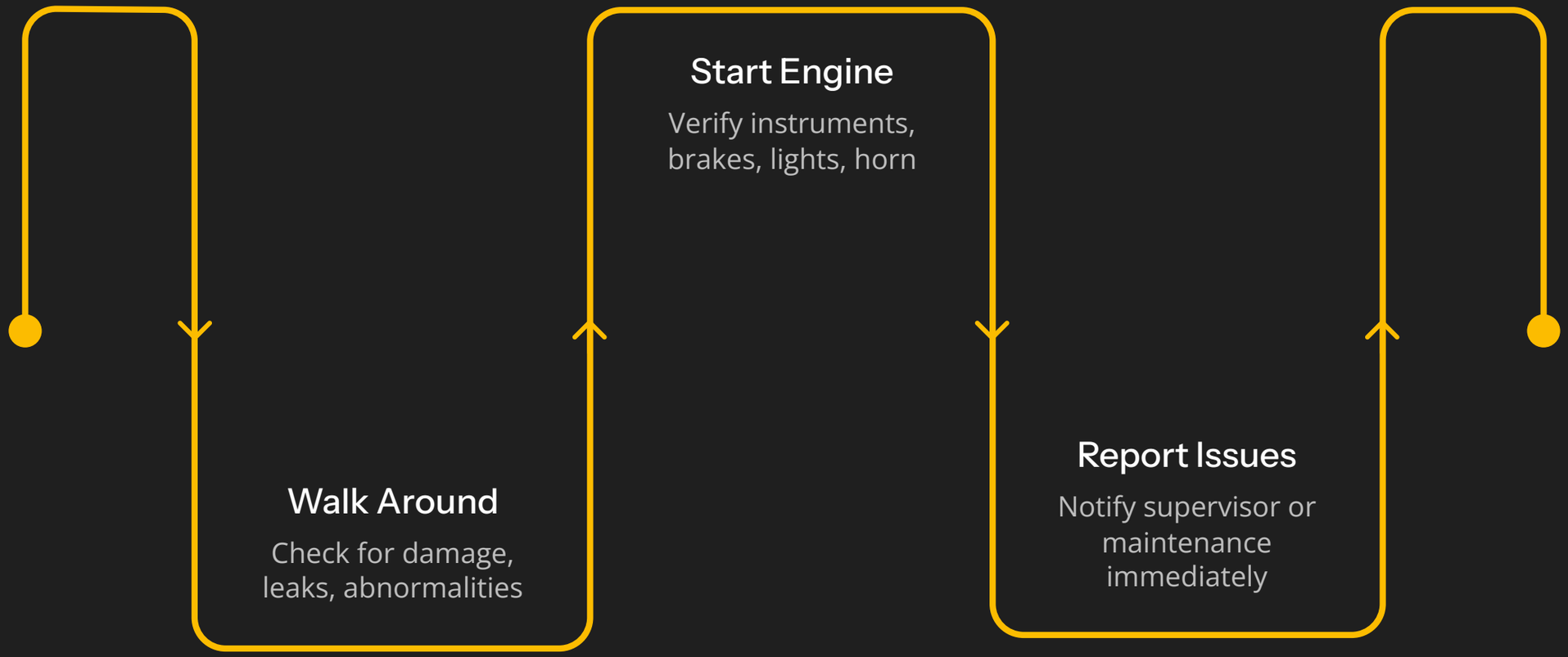
CHAPTER 6

Putting Safety Into Practice

Knowing the rules is not enough — safe forklift operation requires consistent, disciplined adherence to proven operating procedures every single shift. This chapter covers the responsible operating practices that every certified PIT operator must follow.

Rule #1: Inspect Your Equipment Before Every Shift

As the operator, **it is your responsibility** to conduct a thorough daily safety inspection before operating the machine. Do not assume the previous shift left the forklift in safe working condition.



Never operate a forklift that has a known defect or safety issue. Tag it out and report it immediately. Operating faulty equipment puts you and your coworkers at risk.

Forklift Daily Inspection Checklist

Use the following checklist at the start of every shift. Inspect the entire vehicle and record your findings. Report any faults immediately.

WALK-AROUND ITEMS (VISUAL)	OK	NO	N/A
Walk-around inspection (warning decals, capacity plate)			
Forks, locking pins, carriage, mast or boom			
Wheels, tires & lug nuts (condition/pressure)			
Engine (oil level & leaks)			
Transmission (oil level & leaks)			
Radiator (coolant level & leaks)			
Hydraulic tank (oil level & leaks)			
Fuel tank (secure, valve open, fuel level)			
Overhead guard / ROPS (no damage)			
Seatbelt (present and functional)			
STARTUP ITEMS (ENGINE RUNNING)	**OK**	**NO**	**N/A**
Engine sounds normal			
Instruments (normal readings)			
Horn & backup alarm (strobe lights)			
All hydraulic controls (lift, tilt, etc.)			
Brakes (parking & service)			
Steering (all modes)			

Always Buckle Up

Forklift operators **must fasten their seat belt before starting the engine** — every single time, without exception.

In a tip-over event, a properly fastened seat belt is the single most important factor in determining whether an operator survives. The instinct to jump out of a tipping forklift is wrong — **stay in the cab, lean away from the fall**.

OSHA requires seatbelt use on all forklifts equipped with a restraint system. Non-compliance is a citable violation.



⚠ SAFETY RULE

Never Operate With Fluid Leaks

At the start of every shift, inspect the forklift for fuel, oil, and hydraulic fluid leaks. Operating with leaks is strictly prohibited.

- **Hydraulic leaks** can cause sudden, unexpected loss of load control
- **Fuel leaks** create fire and explosion hazards
- **Oil leaks** create slip hazards on warehouse floors

If a leak is discovered, **remove the forklift from service immediately** and notify your supervisor or maintenance team before operating the vehicle again.

NO SMOKING — Ever

Smoking and open flames are strictly prohibited around forklift operations. Sparks or flames near fueling stations and battery recharging areas can ignite flammable vapors and cause catastrophic explosions.

50-Foot Minimum

Sparks or open flames must remain at least **50 feet away** from all fueling stations and battery recharging areas.

Battery Charging Areas

Hydrogen gas emitted during charging is highly flammable. Ensure proper ventilation and prohibit all ignition sources.



⚠ SPEED LIMIT

Max Speed: 5 MPH

Forklifts are engineered to move heavy loads safely — not to race. The maximum safe operating speed in most warehouse environments is **5 miles per hour**.

- Slow down at intersections, blind corners, and pedestrian zones
- Reduce speed further on wet, uneven, or congested floors
- Focus on working **efficiently**, not faster
- Speed is a leading contributing factor in forklift tip-overs

The goal is to complete the task safely — not quickly. Rushing leads to accidents.





⚠ TURNING SAFETY

Do Not Accelerate While Turning

Forklifts have a unique weight distribution — heavy counterweight at the rear, heavy load at the front. This makes them inherently susceptible to tipping, especially during turns.

- Accelerating in a turn shifts the center of gravity dramatically
- Rear-wheel steering means the back of the forklift swings wide
- Always **slow down before turning**, then proceed smoothly
- Never make sudden, sharp steering inputs at speed

⚠ OPERATOR POSITION

Stay Inside the Protective Cage

The overhead guard and operator restraint system are designed to protect the operator — but only if the operator stays fully within the cab. If any part of your body extends beyond the protective cage, **you are not fully protected**. Keep both hands on the wheel and your body inside the cage at all times.



No Passengers — Ever

Your forklift is designed and rated to safely carry **one person only — the operator**. Unauthorized riders face serious risk from sudden stops, turns, or bumps that the operator's restraint system is not designed to protect them from.

The Rule

No hitchhikers. Ever. No exceptions for short rides, convenience, or emergencies.

The Exception

Approved safety platform with 42" guard rails, mid-rail, toe-boards, mast attachment, 7-ft overhead guard, and a personal fall protection harness. See next slide.



SAFETY PLATFORM REQUIREMENTS

Approved Safety Platform Standards

Only one narrow exception exists for carrying an elevated worker: a fully compliant, OSHA-approved safety platform. The platform must include all of the following:

42" High Guard Rails

On all four sides of the platform without exception.

Mid-Rail & 4" Toe-Boards

Mid-rail at the midpoint; toe-boards to prevent foot slippage.

Secured to Mast

Chained or locked to the forklift mast — cannot be free-floating.

7-Ft Overhead Guard + Fall Protection Harness

Mandatory crush protection and personal fall arrest system.





⚠ CRITICAL RULE

Never Leave When Someone Is Elevated

When a worker is in an elevated safety platform, the forklift operator bears full responsibility for their safety. The following rules are absolute:

Stay in the Seat

The operator must remain alert and in the operator's seat at all times while anyone is elevated.

Engine Running

Keep the engine running to maintain hydraulic pressure and platform stability.

Do Not Drive

Never move the forklift to another location while a person is on the platform.

FORK POSITION

Travel With Forks Low to the Ground

Forks are heavy, razor-edged implements that pose a serious hazard if carried at the wrong height. Always maintain the correct travel height — loaded or empty.

Correct Travel Height

4 to 6 inches off the ground — provides clearance over minor floor irregularities while keeping the center of gravity low.

Why It Matters

High forks dramatically raise the center of gravity, increasing tip-over risk and creating a serious strike hazard for pedestrians.



Always Signal Your Approach



Use Your Horn

Sound the horn when entering pedestrian areas, intersections, and areas with limited visibility. Do not assume others can hear or see you.



Blind Corners

Sound the horn **before** approaching any blind corner or doorway. Slow to a crawl before proceeding through the intersection.



Obstructed View: Drive in Reverse

If your load blocks your forward view, **drive in reverse** to maintain a clear line of sight and reduce collision risk.



UNATTENDED EQUIPMENT

Never Leave Your Forklift Unattended

An unsecured, unattended forklift is a serious hazard. Follow these steps every time you leave the operator's seat for any reason:

1. Lower Forks to the Ground
2. Set the Parking Brake
3. Turn Off the Engine
4. Remove the Key if 25+ Feet Away or Out of Sight

CHAPTER 7

Loading Protocol: Safe & Proper Procedures

Improper loading procedures are a leading cause of forklift accidents, tip-overs, and property damage. This chapter covers every critical loading rule — from verifying your forklift's rated capacity to navigating docks and inclines safely.

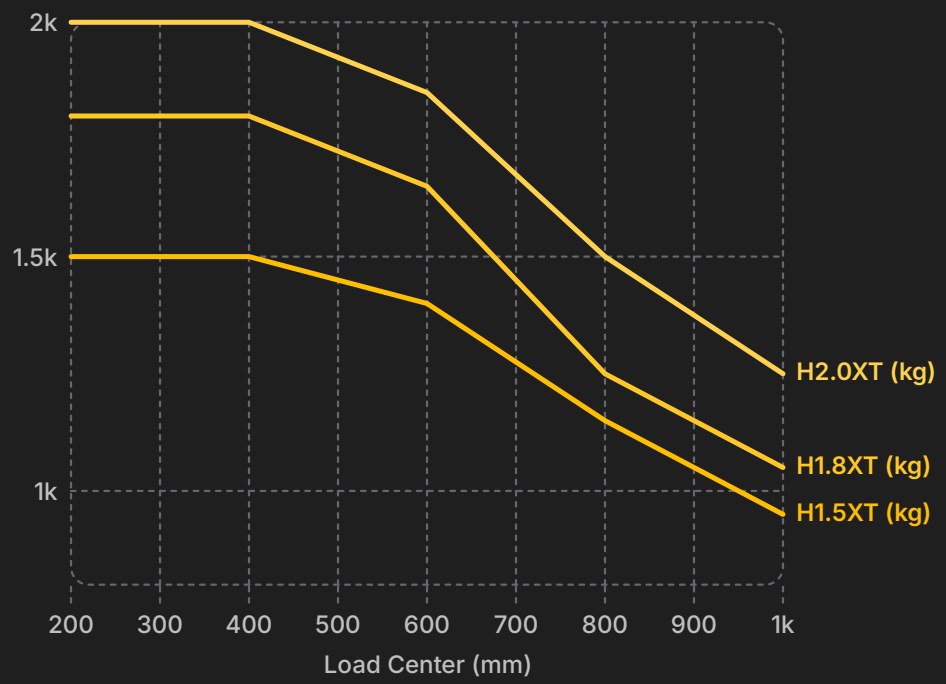


Know Your Forklift's Rated Capacity

Before lifting any load, confirm your forklift is rated to handle its weight and load center. Every forklift has a **manufacturer's data plate** in the operator compartment specifying the maximum lifting capacity.

> **Warning:** Exceeding the rated capacity significantly increases the risk of instability and tip-over — putting you and everyone nearby in serious danger.

A forklift's lifting capacity decreases as the load center distance increases. Always verify you are using the right equipment for the weight **and** size of the load.



As load center distance increases, rated capacity drops significantly. Always verify the data plate for your specific load.



⚠ LOAD SECURITY

Never Move an Unsecured Load

Before transporting any load, ensure it is properly secured, balanced on the forks, and will not shift or spill during movement.

- Use **shrink wrap** or **load straps** to secure unstable or uneven loads
- Center the load evenly across **both forks**
- Ensure load is seated firmly against the carriage backrest
- Never transport a load that obscures your view without driving in reverse

There are no second chances. Always confirm your cargo is secure **before** moving.

Dock Safety: Trailers & Loading Procedures



Chock the Trailer First

Always verify the trailer is properly braked and chocked before entering. Forklift braking force can push a trailer away from the dock — trapping the forklift inside.



Always Use a Dock Plate

Never drive directly from the dock floor to the trailer floor without a dock plate. Keep forks at 4–6 inches when crossing the plate to maintain stability.



Inspect the Trailer Floor

Trailer floors weaken and sag after repeated loading cycles. Always check visually for sagging, soft spots, or holes before driving your forklift inside.

⚠ TRAILER LOADING

Distribute Loads to Prevent Trailer Collapse

Concentrating extremely heavy loads in the center of a trailer can cause the floor to collapse. Proper load distribution is essential every time you load a trailer.

Heavy Loads at the Rear

Position the heaviest loads near the **back of the trailer**, directly above the trailer's axles and wheels for maximum structural support.

Even Distribution

Spread weight evenly side-to-side to prevent trailer lean, which can affect road stability and load security.





INCLINE SAFETY

Always Travel in Reverse on Inclines

When navigating inclines — ramps, dock plates, or uneven terrain — follow these rules strictly to maintain load stability and prevent tip-overs:

1 Tilt the Mast Back

Keep the mast tilted back toward the forklift to shift the load's weight rearward.

2 Travel in Reverse

Drive in reverse when moving up **or** down inclines so the load is always on the "uphill" side.

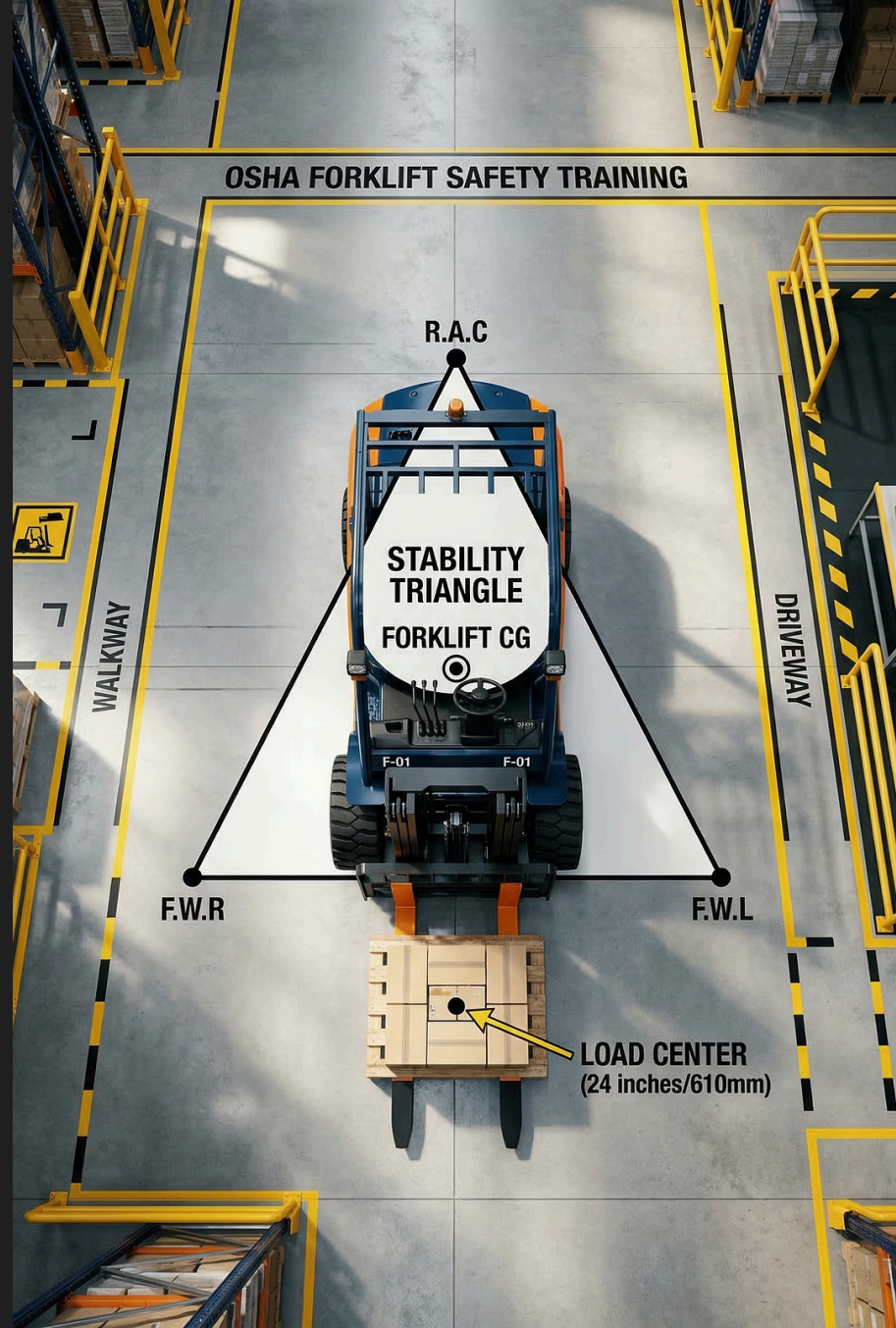
3 Go Slowly

Never rush on inclines. Momentum on a slope dramatically increases tip-over risk.

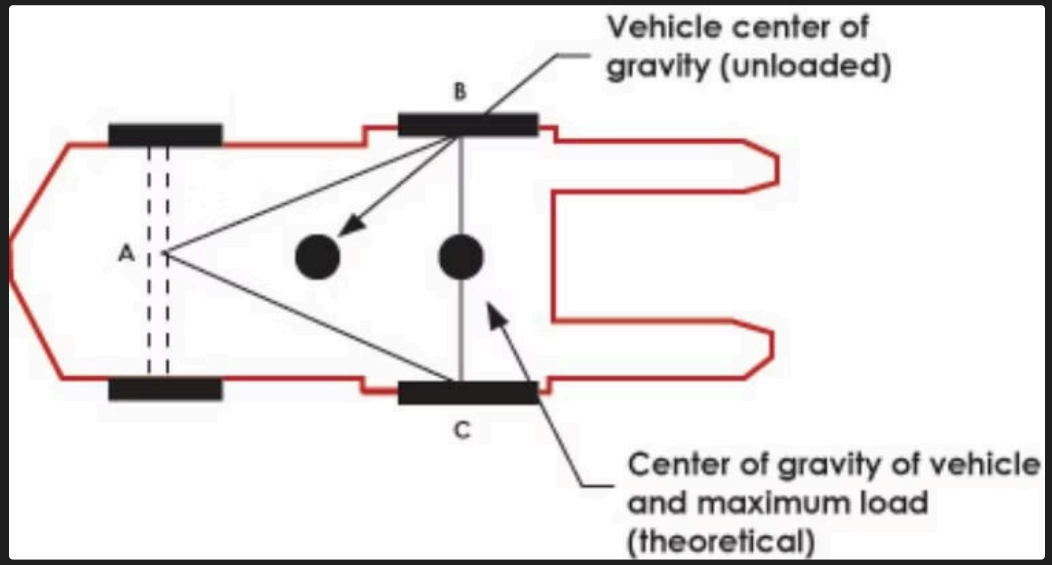
CHAPTER 8

The Stability Triangle

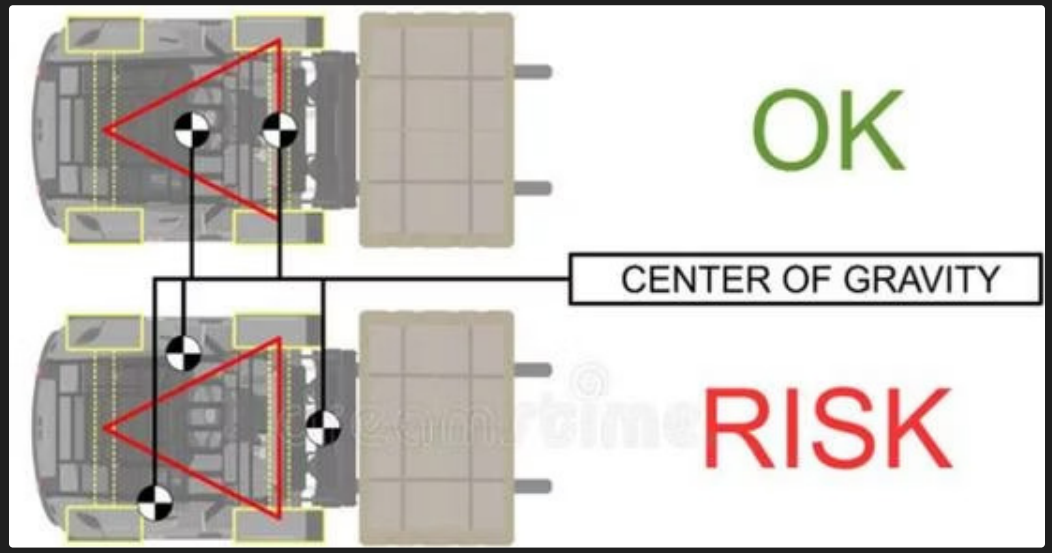
Every counterbalanced forklift has three points of support — the two front wheels and the center of the rear axle. These three points form the **stability triangle**. As long as the combined center of gravity of the forklift and its load remains within this triangle, the forklift remains stable. The moment it shifts outside — the forklift tips.



Center of Gravity: How Loads Shift Stability



Point A = rear axle center. Point B = vehicle CG unloaded. Point C = theoretical CG at maximum load. Combined CG must stay within the stability triangle.



OK: CG inside the triangle — forklift is stable. **RISK:** CG outside the triangle — tip-over is imminent. The load, mast tilt, and driving maneuvers all affect where the CG falls.



⚠ TIP-OVER RISK

Never Transport a Raised Load

Raising the forks elevates the combined center of gravity — moving it dangerously close to (or outside of) the stability triangle. Even moderate braking with a raised load can cause an immediate tip-over.

Actions that dangerously shift the CG:

Tilting Load Forward

Moves CG toward the front — most dangerous position.

Elevating the Load

Raises the CG vertically, dramatically reducing stability.

Speeding Around Turns

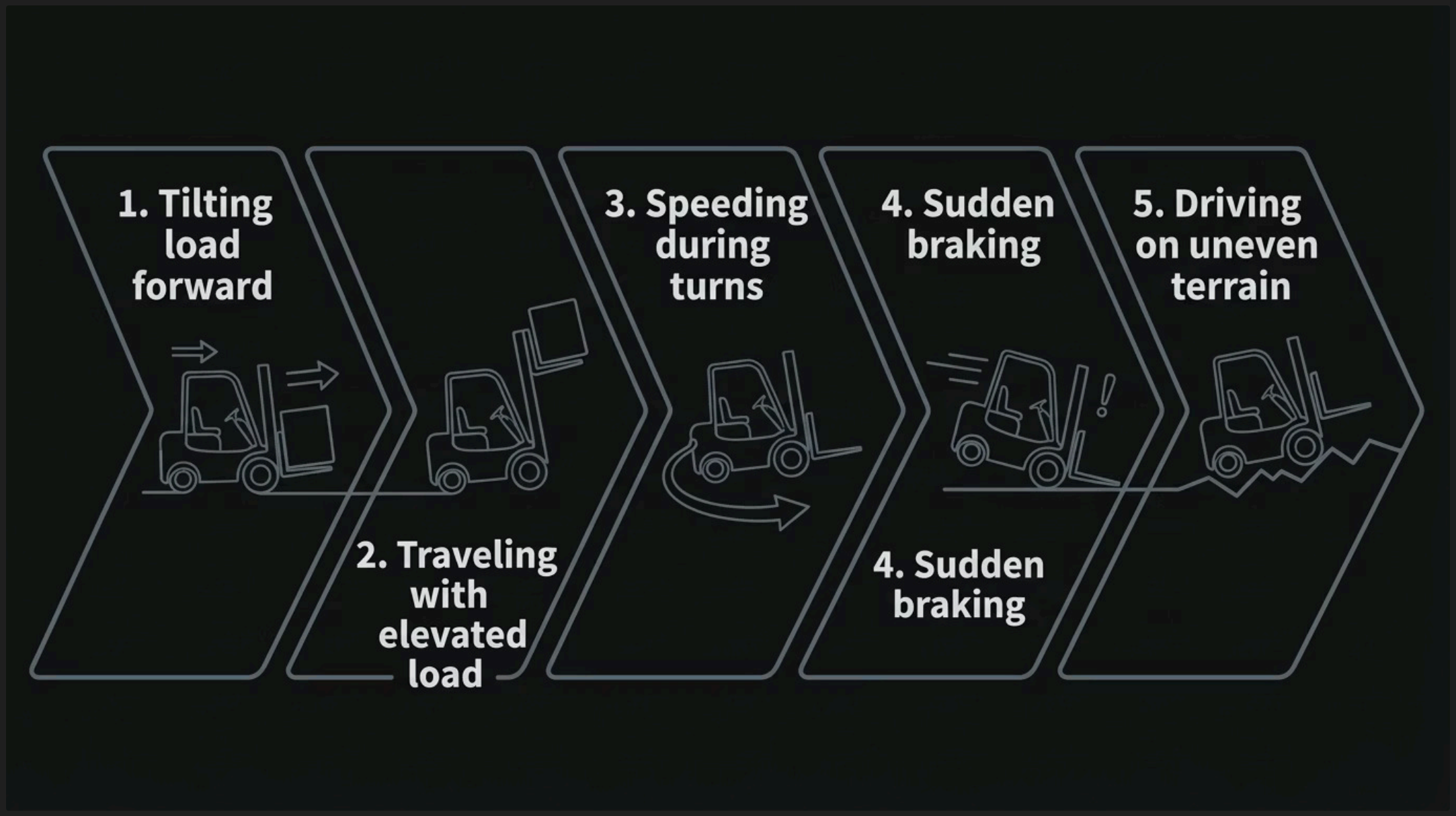
Centrifugal force pushes CG laterally outside the triangle.

Sudden Hard Braking

Momentum surge drives CG forward past the front axle.

Forklifts Are Heavy — But Can Still Tip Over

Despite weighing several tons, forklifts are highly susceptible to tipping. The stability triangle is relatively narrow, and many common operating errors push the center of gravity outside its boundaries within fractions of a second.



If a tip-over begins: **Stay in the cab, grip the steering wheel, brace your feet, and lean away from the fall.** Do NOT attempt to jump out — most tip-over fatalities happen when operators exit the protective cage.



Drive Carefully — Be the Standard

Every certified forklift operator sets the example for safety culture on the floor. The rules in this training are not suggestions — they are life-saving requirements. Apply them every shift, every lift, every time.

Inspect Before Every Shift

Your pre-shift checklist is your first line of defense against preventable accidents.

Operate Within Limits

Never exceed rated capacity. Never bypass the stability triangle. Never rush.

Protect Everyone Around You

Pedestrians, coworkers, and your own wellbeing depend on your discipline every day.

Your Certification: What You Receive



Valid for 3 Years

Upon successful completion of all training requirements, you will receive:

Certificate of Achievement

Official Miramar Forklift Training certificate documenting your OSHA-compliant PIT certification.

Certified Operator Card

A wallet-sized operator license confirming your authorization to operate Powered Industrial Trucks.

Certification must be renewed every **three years** or sooner if unsafe operation is observed, an accident or near-miss occurs, or new equipment types are introduced to your workplace.

FINAL SUMMARY

Key Takeaways: Safety is Your Responsibility

01

Inspect Before Every Shift

Walk around the vehicle, check all systems, and report any defects immediately. Never operate a faulty forklift.

02

Know Your Equipment

Understand your forklift's class, fuel type, tire type, rated capacity, and load center before operating.

03

Follow Operating Rules

Buckle up, stay slow, travel with forks low, signal your approach, and never allow passengers.

04

Load Safely

Verify capacity, secure every load, chock trailers, use dock plates, and keep loads low during transport.

05

Understand Stability

Respect the stability triangle. Never raise loads during transport. Slow down before every turn.



Miramar Forklift Training

OSHA-Compliant Powered Industrial Truck Certification

Safe Operators

Trained. Certified. Authorized.

3-Year Certification

Certificate of Achievement + Operator Card

OSHA Compliant

1910.178(l) and all applicable standards